ESDINet: Efficient Shallow-Deep Interaction Network for Semantic Segmentation of High-Resolution Aerial Images

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***Abstract*— Semantic segmentation of high-resolution remote sensing images is essential in many fields. Nevertheless, in practical applications, constrained by limited computational resources and complex network structures, many advanced models on semantic segmentation often fail to show efficient performance, prompting research on lightweight models. For lightweight semantic segmentation models, the two-branch architecture has been shown to work well in speed and performance. However, such two-branch architectures usually do not utilize enough information for shallow structures to efficiently provide richer multiscale information for the two branches. The lightweight modules it uses are difficult to extract the global context information of the features effectively. Compared with the current advanced semantic segmentation models, lightweight models still have some differences in performance. In order to solve these problems, we propose a new lightweight dual-branch architecture efficient shallow-deep interaction network (ESDINet), which can quickly extract low-level spatial and high-level semantic information of images through the detail branch and semantic branch. Specifically, we have constructed an efficient double-branch structure with shallow and deep different interactions to achieve multiscale information interaction. At the same time, we optimize the semantic branch and propose a new linear attention block to effectively improve the global perception of the semantic branch. We performed extensive experiments and the results show that our model achieves a good balance between segmentation accuracy and inference speed. In particular, ESDINet achieves 82.03% mean intersection over union (mIoU) on the Vaihingen test set, while the proposed model achieves an inference speed of 116 frames/s (FPS) for 512 × 512 inputs on a single NVIDIA GTX 2080Ti GPU.**

***Index Terms*— High-resolution remote sensing image, lightweight, multiscale, semantic segmentation.**

I. Introduction

REMOTE sensing semantic segmentation is a fundamental computer vision task that aims to assign semantic labels to each pixel point on a remotely sensed image. With the continuous development of this technology, it has wide applications in many scenarios, such as environmental monitoring [1], [2], crop cover and type analysis [3], and urban spatial building classification and land use analysis [4], [5]. The advancement of these applications has further driven researchers to explore even more efficient models, particularly when faced with resource-constrained devices or scenarios requiring rapid computation.

In the past, semantic segmentation methods were usually based on threshold segmentation, edge detection, or traditional machine learning [6], [7], [8], [9], [10], which often rely heavily on manual features and can only extract part of the limited low-level information, with insufficient extraction capability for high-level semantic information and cannot adequately capture the relevant context between image pixels. Since the rise of deep learning [11], [12], [13], especially the proposal of full convolutional network (FCN) [14], the first semantic segmentation model for end-to-end learning, a large number of semantic segmentation models based on convolutional neural networks, such as SegNet [15], UNet [16] pyramid scene parsing network (PSPNet) [17], and DeepLabV3 series [18], [19], have emerged, showing performance far beyond that of traditional algorithms. They extract finer and different scale spatial features through specific encoding–decoding structures or some pyramidal multiscale modules, which make the segmentation results further sophisticated. The success of these models shows that rich spatial detail information and interpixel contextual correlation are extremely important in semantic segmentation tasks, and the models need to have both multiscale perception and large sensory field extraction capabilities. At the same time, despite the good performance achieved by these models, the pursuit of accuracy in module and structural design makes many calculations very expensive and difficult to apply directly in real situations characterized by limited device resources or the need for fast computations.

Many lightweight semantic segmentation models, such as ENet [20] and ICNet [21], have been proposed to address this problem. They greatly reduce the computational cost of the network by using lightweight backbone [22], [23] and deeply separable convolutional blocks or by targeting a restricted image resolution size, allowing the model to maintain a certain accuracy while ensuring real-time inference. In particular, the bilateral network structure was suggested by Bilateral segmentation network (BiSeNet) [24]. It involved three fairly broad 3 × 3 convolutional layers for the spatial

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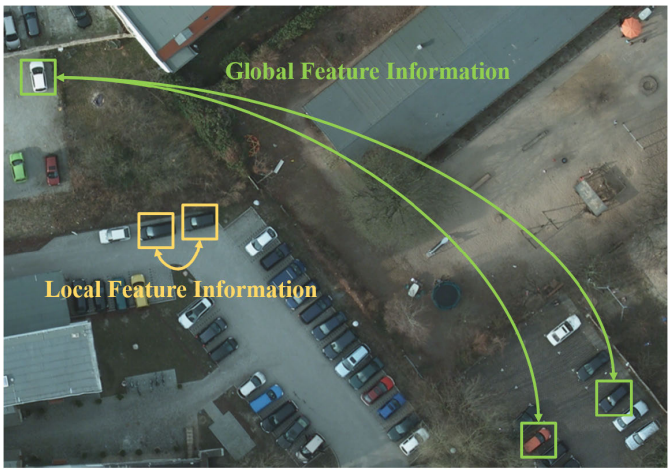


Fig. 1. Illustration of the difference between local feature information and global feature information.

branch, focusing on spatial details. In addition, a lightweight backbone network was employed for the context branch and targeting contextual information. A fusion module combined low-level details and high-level semantics. This architecture became a competitive real-time approach during that time. However, in the face of high-resolution remote sensing images, which contain rich feature details with many small target classes, the use of lightweight models still faces the problem of limited segmentation accuracy.

On the one hand, the simple convolutional modules used by lightweight models tend to perform only a certain degree of feature extraction of local information, and the perceptual field is difficult to raise to the global level, which limits the network's ability to extract context. With the success of vision transformer (ViT) on vision tasks, its softmax attention module has emerged as a solution to improve the global information extraction capability of the model as a method capable of capturing long-range dependencies (as shown in Fig. 1). However, the computational cost of softmax attention grows quadratically with the input resolution and is expensive for semantic segmentation tasks with high resolution. To alleviate this problem, Katharopoulos et al. [25] addressed linear attention and reduced the complexity of the self-attention mechanism from $O(N^2)$ to $O(N)$, maintaining high accuracy while increasing the computational speed. On the other hand, the two-branch structure represented by BiSeNet tends to fuse information only in the deep part of the network, and there is no interaction operation for the information in the shallow part of the network, making a large amount of multiscale information in the two-branch wasted to some extent. And, the utilization of this multiscale information is important for the task of high-resolution segmentation and largely influences the fineness of the final segmentation.

To this end, we propose a lightweight and efficient semantic segmentation network structure, as shown in Fig. 2(c). The network consists of a wide channel, a detail branch for spatial detail extraction, and another narrow channel, a semantic branch for high-level semantic extraction. Different information interactions are used between the dual branches in the shallow and deep parts of the network to improve the utilization of spatial multiscale feature information at different levels of the network, achieve more effective information fusion, and improve model performance at a negligible computational cost. In addition, we construct our linear moving attention based on the linear attention mechanism (LAM) and combine it with ResNet-18 to implement a lightweight convolutional neural network (CNN) hybrid transformer structure that can effectively capture both local details and global features. Throughout the process, due to the lightweight nature of the modules we use, the proposed efficient shallow-deep interaction network (ESDINet) is able to perform inference at 116 FPS in real-time on a 2080Ti. By comparing with several mainstream methods on the Vaihingen and Potsdam datasets, the experimental results show that our proposed lightweight ESDINet has a good balance of accuracy and speed.

Our main contributions are summarized as follows.

1) We propose a new semantic segmentation model structure that introduces different information interactions in the shallow and deep parts of the network. It provides richer complementary information for detail and semantic branches and improves the overall segmentation performance of the model.
2) For the shallow representation of the network, we build simple and efficient shallow interaction (SI) modules to improve the information complementarity of details and semantic branches. For a deeper representation of the network, we build lightweight fusion modules capable of capturing high-level feature channels and location information.
3) We introduce an LAM to build lightweight linear mobile attention (LMA), which is combined with local convolution to achieve a lightweight hybrid architecture that can capture local detail and global feature information with low computational complexity.

II. Related Works

A. Classical Semantic Segmentation Model

Since the impressive performance of the FCN [14] approach on various benchmarks, improved models based on convolutional neural networks started to be used to solve segmentation tasks in various natural images. SegNet [15] modified the method of upsampling operation based on FCN to recover high-resolution feature maps using an encoder–decoder structure. UNet [16] proposed a more complex U-shaped encoder–decoder structure, which makes the decoder high-resolution feature information recovered more finely by top–down and lateral residual connections; RefineNet [26] refined the prediction results by adding a multipath refinement structure; DeepLabV3 series [18], [19] introduced atrous spatial pyramid pooling module to capture multiscale contexts. PSPNet [17] designed a multiscale pyramid pooling module in order to capture local and global contextual information on the spreading backbone. Dual attention network (DANet) [27] is based on a self-attention mechanism, which captures rich contextual dependencies to accomplish the scene segmentation task. PSANet [28] introduces a pointwise spatial attention network to relax the constraints of local neighborhoods, obtaining more global and rich contextual information to

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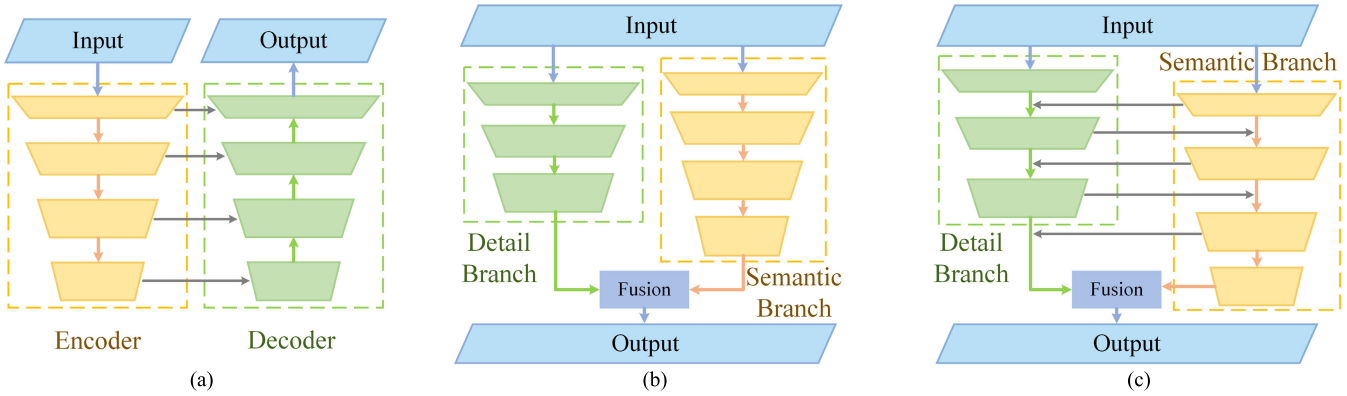


Fig. 2. Illustration of different network architectures. (a) Encoder–decoder structure like UNet. (b) Two-branch structure like BiSeNet. (c) ESDINet structure like the one we propose.

enhance the model's performance. In the field of remote sensing, many deep learning-based models have also been proposed for the high-resolution properties of remote sensing images. Fu et al. [29] used Atrous convolution to optimize the FCN model and used conditional random field (CRF) [30] to postprocess the segmented data. EdgeFCN proposed by He et al. [31] used edge information collected by a holistic nested edge detection (HED) network to calibrate the FCN segmentation results. The relational context-aware fully convolutional network (RA-FCN) proposed by Mou et al. [32] uses a spatial relationship module and a channel relationship module to learn and reason about the global relationship between any two spatial locations or feature maps. The class-guided feature decoupling network (CGFDN) [33] approach improves the segmentation accuracy by extracting more contextual information from the aerial images.

It can be found that low-level details and high-level semantics can be learned by extending the backbone, encoder–decoder structure, or multiscale modules, which can effectively improve the model segmentation accuracy. However, due to the high-resolution nature and complex network design, these methods often require relatively large computational costs. Achieving faster inference requires targeted and lightweight improvements. Meanwhile, the semantic segmentation task-oriented to pixel-level classification needs to further improve the global information extraction ability of the model due to the insufficient perceptual field of the convolutional module.

B. Attentional Mechanisms

The attentional mechanism is a technique that simulates the mechanism of human observation behavior. It has been shown to be effective in improving the performance of models for information extraction and enhancing attention to useful features. As in the squeeze-and-excitation network (SE) module [34], a global average pooling layer and a linear layer are used to compute the scaling factor for each channel and reweight operations on information channel features. The convolutional block attention module (CBAM) [35], by generating attentional feature map information in both channel and space dimensions, performs adaptive feature correction for both channel and space. Coordinate attention [36] proposes a lightweight network attention to enhance the representation of objects of interest by embedding location information into the channel attention.

Recently, inspired by the success of Transformer [37] in natural language processing (NLP), ViT [38] has received a lot of attention and is widely used in various computer vision tasks. One of the main dependencies is the softmax attention module, which is able to model the interaction between each pair of tokens in the feature map to extract global information. However, it has a secondary computational complexity in terms of input resolution, for which: 1) it is to reduce the size of the key/value tensor [39], [40], which makes the number of values involved in the computation reduced, substantially lowering the computational cost and 2) it is to restrict the softmax attention to a fixed size, such as the shift window used by Swin Transformer [41]. Although these methods reduce the computational cost of softmax attention. However, they sacrifice the advantage of ViT in global feature extraction. Based on this foundation, research has surfaced focusing on enhancing the efficiency of ViT. One notable approach involves devising a hybrid architecture that integrates CNN modules with the Transformer framework. For instance, MobileViT [42] employs a fusion of CNN and ViT strengths, culminating in the creation of a streamlined and efficacious hybrid model. Another innovation, namely, TopFormer [43] and amalgamates stacked MobileNet Blocks [44] with a multiscale Transformer setup, thereby striking an optimal balance between precision and efficiency. Another is to reduce the computational complexity of softmax attention [45], [46], where Katharopoulos et al. [25] conducted a study of linear attention, which reduced the softmax attention complexity from $O(N^2)$ to $O(N)$, resulting in a great reduction in computational cost. We introduce a lightweight and efficient linear mechanism attention (LMA) block. By synergistically integrating this LMA block with convolutional layers, we devise an efficient hybrid architecture. This novel architecture inherently encompasses the benefits of both global feature perception and localized detail extraction. Consequently, it significantly amplifies the feature extraction capacity within the semantic branch.

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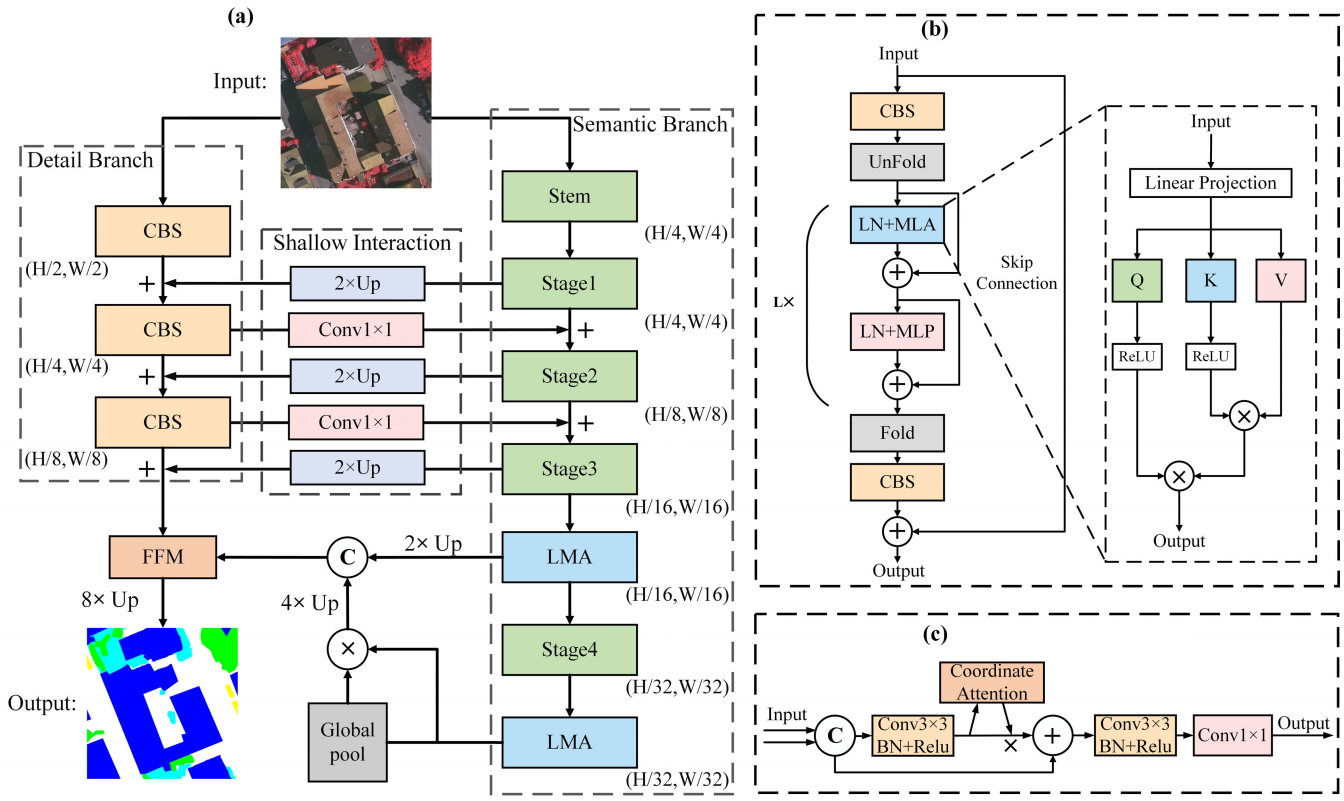


Fig. 3. Network framework of ESDINet and the details of each module. (a) Overall architecture of the network, includes a semantic branch, a detail branch, and the shallow and deep fusion modules. (b) LMA. (c) FFM. CBS means Conv1 × 1/Conv3 × 3 + BatchNorm(BN) + SiLU. $N\times$UP means N times interpolation upsampling. C represents the Concatenate operation.

C. Effective Lightweight Design

Lightweight and efficient design occupies an important significance in the real-time inference of models and has been widely valued by computer vision tasks. The use of depth-separable convolution in MobileNetV1 [44] drastically reduces the number of parameters, and the computational amount of ordinary convolution. ResNet [47] uses residual connections to reduce the network degradation problem and improve the model performance. MobileNetV2 [48] proposed the inverted residual with linear bottleneck structure to maintain the accuracy while reducing the computational cost. ShuffleNet [49], [50] uses a moderate amount of group convolution to maintain a balance between accuracy and speed. Based on these generic and effective designs, the construction of a lightweight semantic segmentation model mainly consists of: 1) lightweight backbone networks and modules and 2) simple and efficient structures. ENet [20] improves the inference speed of the model by greatly reducing the input resolution size. ICNet [21] improves the performance of lightweight models by multiscale image cascading. BiSeNetV1 [24] and BiSeNetV2 [51] propose two streaming paths to achieve a balance of accuracy and speed for low-level detail and high-level contextual information, respectively. Fast-SCNN [52] is based on a two-way architecture that reduces computation by sharing shallow parameters of different branches.

Some models using lightweight schemes have also been proposed for remote sensing segmentation tasks, and they provide some degree of improvement in the number of parameters and computational effort of the models. Efficient-Transformer [53] proposes explicit and implicit edge enhancement methods to deal with object edge problems based on Swin Transformer. MAResU-Net [54] introduces an LAM and reconfigures the U-Net skip connections to obtain more refined segmentation results. Cross fusion net (CF-Net) [55] constructs lightweight networks to capture more fine-grained information through channel attention refinement blocks and cross-fusion blocks. MANet [56] improves the ability of the model to extract global contextual information based on linear kernel attention and channel attention. Attentive bilateral contextual network (ABCNet) [57] is based on a two-branch architecture and improves the model performance through the constructed linear attention enhancement module. Our proposed ESDINet structure, responding to different characteristics of the shallow and deep layers, respectively, is a light and effective interaction strategy that fully utilizes different scale information of the two branches and makes up for the lack of interaction of the shallow information in the two-branch structure, effectively improving the segmentation performance of the whole network.

III. Method

In this section, we will elaborate on the core concept of the proposed ESDINet, and the whole architecture can be decomposed into three parts: detail branch, semantic branch, and interaction module. The overall framework diagram of the ESDINet model proposed in this article is shown in Fig 3.

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A. Detail Branch

In the semantic segmentation task, an encoder–decoder structure, i.e., downsampling–upsampling, is usually used to achieve the full process of segmentation. However, the process of repeated downsampling leads to the loss of some of the original spatial information, which is difficult to be fully recovered by upsampling; therefore, it is crucial to ensure the rich spatial detail information for the final semantic segmentation accuracy. Similar to BiSeNet [24], our proposed ESDINet also uses the detail branch to capture low-level spatial information. Specifically, we use a simple shallow structure to extract rich spatial low-level information, which focuses more on the underlying detailed features compared with the semantic branch. We use three layers of feature extraction blocks constructed from the 3×3 Conv-BN-SiLU operator (CBS) and increase the number of channels of feature output layer-by-layer (64, 128, and 256).

With this lightweight and shallow design, the three layers of the detail branch extract 1/2, 1/4, and 1/8 size feature maps of the original image, and each feature representation is characterized by wide channel and high resolution, containing rich spatial detail information.

B. Semantic Branch

In order to provide a sufficiently large receptive field, we designed a semantic branch that is deeper and more complex than a detail branch. In contrast to the direct use of a lightweight backbone to extract contextual information in BiSeNet, we use a lightweight hybrid structure of CNN and Transformer as semantic branches. Specifically, we combine the convolutional layer of ResNet-18 with the proposed LMA Module to design a more efficient lightweight hybrid model.

1) Linear Attention Mechanism: ViT [38] was shown to achieve significant performance in many tasks in vision, where the heavily used self-attention mechanism is considered to be one of the core parts. Using self-attention enables efficient remote interaction and capturing global information compared to local convolutional blocks. However, self-attention has a secondary computational complexity and incurs high computational cost and memory consumption in high-resolution tasks. Recalling the attention strategy, for an input $x \in \mathbb{R}^{N\times d}$, the softmax attention can be written as follows:

$$O_i = \sum_{j=1}^{N} \frac{\mathrm{Sim}\left(Q_i, K_j\right)}{\sum_{j=1}^{N} \mathrm{Sim}\left(Q_i, K_j\right)} V_j \tag{1}$$

where $Q = xW_q$, $K = xW_k$, and $V = xW_v$, $W_q/W_k/W_v \in \mathbb{R}^{d\times M}$ is a learnable matrix, O_i denotes the ith row of matrix O, and $\mathrm{Sim}(\cdot,\cdot)$ is the similarity function between the metric queries. When $\mathrm{Sim}(Q,K) = \exp((QK^{\mathrm{T}}/(D)^{1/2}))$, (1) represents the softmax attention

$$A = \mathrm{Sim}(Q_i, K_j). \tag{2}$$

Equation (2) represents the similarity attention map obtained by computing Q and K^{T}. It is sufficient to ensure that the attention matrix A is nonnegative, and we can choose any similarity function Sim to compute the attention matrix A. To maintain a linear computational budget, one solution is to use decomposable similarity functions as in linear attention [25] such that

$$\mathrm{Sim}\left(Q_i, K_j\right) = \phi(Q_i)\phi\left(K_j\right)^{\mathrm{T}} \tag{3}$$

where ϕ is the kernel function that maps queries and keys to their hidden representations. Equation (1) can then be rewritten in the form of a kernel function as follows:

$$O_i = \frac{\sum_{j=1}^{N} \phi(Q_i)^{\mathrm{T}}\phi\left(K_j\right)V_j}{\sum_{j=1}^{N} \phi(Q_i)^{\mathrm{T}}\phi\left(K_j\right)}. \tag{4}$$

After that, the attention operation in linear complexity is implemented by the matrix product property

$$\left(\phi(Q)\phi(K)^{\mathrm{T}}\right)V = \phi(Q)\left(\phi(K)^{\mathrm{T}}V\right). \tag{5}$$

That is, the combination property of matrix multiplication can be simplified by

$$O_i = \frac{\phi(Q_i)^{\mathrm{T}}\sum_{j=1}^{N} \phi\left(K_j\right)V_j^{\mathrm{T}}}{\phi(Q_i)^{\mathrm{T}}\sum_{j=1}^{N} \phi\left(K_j\right)}. \tag{6}$$

By the above operation of rearranging the dot product, we obtain (6) with linear complexity. Meanwhile, we choose ϕ as the rectified linear unit (ReLU) function and fine-tune equation (6) due to its nonnegative and lightweight property, and the final linear attention is calculated as shown in the following equation:

$$O_i = \frac{\mathrm{ReLU}(Q_i)\sum_{j=1}^{N} \mathrm{ReLU}\left(K_j\right)^{\mathrm{T}}V_j}{\mathrm{ReLU}(Q_i)\sum_{j=1}^{N} \mathrm{ReLU}\left(K_j\right)^{\mathrm{T}}}. \tag{7}$$

We construct the LMA block as shown in Fig. 3(b) based on linear attention. Given an input $\mathbf{Z}_{\mathrm{in}} \in \mathbb{R}^{H\times W\times C}$, where H and W are the height and width of the feature map, respectively, and C is the number of channels. At the beginning of the linear attention branch, the channels are first compressed by 1×1 CBS, and the output is $\mathbf{Z}_d \in \mathbb{R}^{H\times W\times D}$, where D is the compression number we set, the two blocks are set to 96 and 192, respectively, which can reduce the computational cost to some extent. Inspired by Mehta and Rastegari [42], we spread the features after the unfold operation into $\mathbf{Z} \in \mathbb{R}^{N\times P\times D}$, where P is the token dimension (patchsize^2) and N is the number of token blocks $[(HW/P)]$. Compared with the ViT operation ($\mathbf{X}_{\mathrm{in}} \in \mathbb{R}^{H\times W\times D} \rightarrow \mathbf{X} \in \mathbb{R}^{N\times(P\cdot D)}$), here channel D is not merged with P so that the spatial order of pixels within each patch is not lost and is more applicable to feature maps with multiscale inputs. Fig. 4 shows the specific operating instructions. Then, we perform the computation of the attention graph by multiheaded linear attention (MLA) to

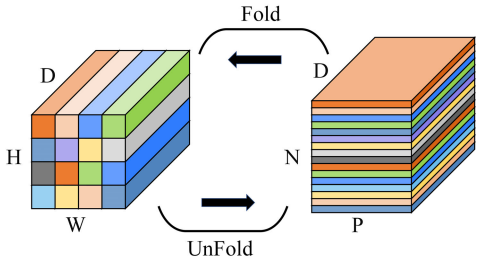


Fig. 4. Instructions for fold and unfold operations, spatial order characteristics remain unchanged.

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capture the global information with $O(N)$ complexity, where the number of multi heads we set to 4, and the length (L) of both LMA blocks is set to 2. After the multilayer perceptron (MLP) layer, $\mathbf{Z}_\ell \in \mathbb{R}^{N\times P\times D}$ are then folded into $\mathbf{Z}_f \in \mathbb{R}^{H\times W\times D}$, and information are integrated by 1×1 CBS and up-dimensioned back to the original $\hat{\mathbf{Z}} \in \mathbb{R}^{H\times W\times C}$, the whole process does not change the original input size. In addition, since the unfold operation makes the token spatially sequential, our module does not need to introduce additional positional encoding information. A standard linear branch is calculated as follows:

$$\begin{aligned}
\mathbf{Z} &= \text{UnFold}(\text{CBS}(\mathbf{Z}_{\text{in}}), && \mathbf{Z} \in \mathbb{R}^{N\times P\times D}\\
\mathbf{Z}'_\ell &= \text{MLA}(\text{LN}(\mathbf{Z}_{\ell-1})) + \mathbf{Z}_{\ell-1}, && \ell = 1\ldots L\\
\mathbf{Z}_\ell &= \text{MLP}\big(\text{LN}\big(\mathbf{Z}'_\ell\big)\big) + \mathbf{Z}'_\ell, && \ell = 1\ldots L\\
\hat{\mathbf{Z}} &= \text{CBS}(\text{Fold}(\mathbf{Z}_\ell)), && \hat{\mathbf{Z}} \in \mathbb{R}^{H\times W\times C}. \qquad (8)
\end{aligned}$$

In general, the linear attention has some advantages in terms of computational cost; however, the accuracy performance may be slightly lower compared with softmax attention. We believe that the linear attention block requires additional auxiliary enhancements, and thus, the additional enhancements should not introduce excessive computational effort. Compared with complex operations, such as convolution, we find that residual concatenation is a very simple and effective method. Here, we introduce the residual branch as an auxiliary summation, which is equivalent for providing more paths for feature learning, and also bring in the convolutional local feature information from the upper layer to assist the linear attention block for feature extraction and improve the overall performance of the model. The output is represented by the following equation:

$$\text{LMA}(x) = \text{LB}(x) + \text{RB}(x) \qquad (9)$$

where LB and RB represent the linear attention branch and the residual branch, respectively.

2) Hybrid Structure: Numerous studies have shown that the pure ViT structure, although capable of extracting global properties, is not suitable for direct use in high-resolution tasks due to its high computational cost. While in the lightweight domain, convolution is often more advantageous in terms of performance and model complexity, considering a hybrid model of CNN and Transformer can combine the features of both. We use ResNet-18 as the backbone network of the convolutional layer for local feature information extraction. It first goes through a stem layer consisting of convolution and maximum pooling, which down samples the original image by a factor of 4. Then, there are four stages consisting of a certain proportional number of residual blocks. each stage performs a different level of depth feature extraction on the data, providing a multiscale feature input. We use the corresponding stage structure of ResNet-18 for feature extraction in Stages 1 and Stage 2 in the semantic branch without adding any additional module, and the output resolution is 1/4 and 1/8 of the input image, respectively, keeping the original semantic feature information at a shallow level. In Stage 3 and Stage 4 of the network, the features have more advanced semantic information as they undergo more downsampling of the local convolutional blocks. We fuse the corresponding stage layer with the proposed LMA block. Since the features are introduced at a lower resolution (32×32 and 16×16) in the later stages of the network, our LMA block brings a low amount of parameters throughout the process. As shown in the semantic branch of Fig. 3(a), a hybrid architecture of CNN and Transformer is formed with both local perception and global information semantic extraction capabilities.

C. Interaction Structure

As with the common two-branch architecture model, our proposed detail and semantic branches are able to extract low-level detail information and high-level semantic features of the full graph, respectively. Usually, in a two-branch network, the shallow features are usually not subjected to cross-fusion operations, reducing the utilization of information. However, the shallow information feature representations of different branches also contain complementary information. In order to more fully exploit the different feature representations, we propose a lightweight and efficient interaction mechanism that combines different features of semantic and detail branches for different interactions. In the shallow structure, detail information is mainly extracted by three layers of convolution in the detail branch, and low-level features are captured by the convolutional layers of Stages 1-3 in the semantic branch. Since they all perform fewer feature extraction operations, the output feature space resolution is relatively large, and using complex interaction modules would make the model computation very expensive.

1) Shallow Interaction: Considering the impact of performance and accuracy, we adopt a simple interaction structure to achieve the fusion of shallow information of two branches, as shown in the SI in Fig. 3(a). Specifically, for each layer output $\mathbf{F}_i^d$ of the detail branch and the shallow output $\mathbf{F}_j^s$ of the semantic branch convolution, we up sample the feature layer $\mathbf{F}_i^d$ in the semantic branch with the corresponding resolution half of it first so that $\mathbf{F}_i^d$ and $\mathbf{F}_j^s$ have the same spatial resolution, and then sum $\mathbf{F}_i^d$ and $\mathbf{F}_j^s$ as the next layer input to the detail branch. And for the shallow output $\mathbf{F}_j^s$ of the semantic branch, we perform channel aggregation by 1×1 convolution of features $\mathbf{F}_i^d$ with the same resolution in the detail branch, so that the number of channels of $\mathbf{F}_i^d$ is compressed to be the same as $\mathbf{F}_j^s$, and then perform summation as the input of the next layer of the semantic branch. The whole process is shown in the following equations:

$$\mathbf{F}_i^d = \mathbf{F}_{i-1}^d + \text{Up}(\mathbf{F}_j^s) \qquad (10)$$

$$\mathbf{F}_j^s = \mathbf{F}_{j-1}^s + C_{1\times 1}(\mathbf{F}_i^d). \qquad (11)$$

The computational cost of the whole process is small because of the simple and effective interaction mechanism. Through the SI, the detail branch obtains high-level semantic feature information in the semantic branch, and the semantic branch is complemented by the shallow spatial details of the detail branch. Fig. 5 illustrates the visual interpretation. The visualization shows that introducing SIs improves the spatial

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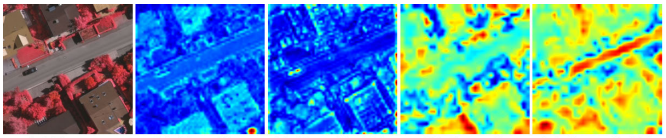


Fig. 5. Visual interpretation of the introduction of SIs. Details denote the output of the last layer of the detail branch, Stage 3 denotes the output of Stage 3 layer with LMA in the semantic branch, and those with -SI denote the result of introducing SIs. (a) Input. (b) Details. (c) Details-SI. (d) Stage 3. (e) Stage 3-SI.

information refinement of the detail branch and enriches the local details of high-level features in the semantic branch.

2) Deep Interaction: In the deep structure of the network (i.e., the last layer of output detail branches, Stage 3, and Stage 4 include semantic branches of linear attention blocks), we first perform a channel rescreening operation on Stage 4 and then concatenate the features of Stage 3 and Stage 4 by linear interpolation upsampling. Then, we designed a feature fusion module (FFM) for integrating features, as shown in Fig. 3(c). Since there are more representational differences between the detail and semantic branches in the deeper features, refinement is needed to fuse these features. We first join the features of two branches together, perform channel aggregation using the 3×3 Conv-BN-ReLU operator (CBR), and make the channels drop to 1/32 of the original. Then, the coordinate attention [36] is introduced to capture cross-channel information and location-sensitive information by coordinating attention to increase the weight of important information of features for feature rescreening. After residual summation, feature aggregation is completed. We perform channel downsampling using the 3×3 CBR and 1×1 convolution of the splitting head. The whole process can be denoted

$$\mathbf{F}_{\text{cat}} = \text{Concat}(\mathbf{F}_l^d,\ \mathbf{F}_l^s) \tag{12}$$

$$\mathbf{F}_{\text{out}} = \mathbf{F}_{\text{cat}} + f_{\text{ca}}(\text{C}_{3\times3}\,\text{BR}(\mathbf{F}_{\text{cat}})) \tag{13}$$

where $\mathbf{F}_l^d$ and $\mathbf{F}_l^s$ are the last layer outputs of the two branches, respectively, Concat$(\cdot)$ represents the concatenate operation, $f_{\text{ca}}(\cdot)$ represents the coordinate attention, and $C_{3\times3}$ BR represents Conv 3×3 BN-ReLU operation.

After deep feature fusion in the FFM layer, we use an $8\times$ bilinear interpolation upsampling operation to output the results.

IV. Experiment

A. Dataset

We assessed our ESDINet's effectiveness on the ISPRS Vahingen, ISPRS Potsdam, and LoveDA [58] datasets, and also conducted comparisons with other semantic segmentation networks. There are two ground truth options available in ISPRS: those with erosion boundaries and those without erosion boundaries. We chose the ground truth with eroded boundaries in the course of all experiments.

The Vaihingen dataset consists of 33 images with an average size of 2494×2064 pixels. True orthophoto (TOP) in the near-infrared response (NIR), red and green bands, and digital surface models (DSMs) in individual bands are provided in the dataset, all of which are sampled at a distance of 9 cm from the ground. In the Vaihingen data, there are 16 images in the training set and 17 images in the test set. We used IDs: 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38 for testing, ID: 30 for validation, and the remaining 15 images for training. In addition, we did not use DSM throughout the experiment.

The Potsdam dataset contains 38 fine-resolution images of size 6000×6000 pixels, and the dataset provides TOP for NIR, red, green, and blue channels as well as DSM and normalized DSM (NDSM) with a ground sampling distance of 5 cm. The six feature classes are the same as in the Vaihingen data, namely, impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. In the Potsdam data, there are 24 images in the training set and 14 images in the test set. Also consistent with [56], we used IDs: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13 for testing, IDs: 2_10 for validation, and ID: 2_10 except for the image named 7_10 with the wrong annotation for the remaining 22 images for training. Here again, we did not use DSM and only selected images from the red, green, and blue channels for the same experiment.

The LoveDA dataset consists of two scenes (urban and rural) collected from three cities in China (Nanjing, Changzhou, and Wuhan). It comprises a total of 5987 high-resolution optical remote sensing images with dimensions of 1024×1024 pixels and a ground sampling distance of 0.3 m. These images encompass seven land cover categories, including buildings, roads, water bodies, barren land, forests, agriculture, and background. The dataset poses significant challenges due to its complex background samples, multiscale objects, and inconsistent categories.

B. Experimental Setup

All models in the experiments were trained and tested on a single NVIDIA GeForce RTX 2080Ti using the PyTorch framework with batch size set to 24. To reduce the effect of the learning rate, we use AdamW [59] as an optimizer with a learning rate set to 1e−3 and a weight decay value of 2.5e−4. A cosine learning strategy was also used to adjust the learning rate. The maximum training iteration period is set to 1000 epochs. In the training phase, we used basic data augmentation, i.e., random horizontal flipping with vertical flipping, and random scaling with 0.5, 0.75, 1.0, 1.25, 1.5, 1.75, and 2.0 random scale for a cut block of dataset size 512×512 to improve the generalization ability of the model. The ResNet-18 backbone of the model was initialized using pretraining on ImageNet-1k [60], and the rest of the model was used with the default initialization of PyTorch. During the testing phase, we used a test time augmentation (TTA) that included horizontal flips as well as vertical flips. On the loss function, we used the generic cross-entropy loss

$$L_{ce} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log(p_{ij}) \tag{14}$$

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TABLE I

ABLATION EXPERIMENT OF THE PROPOSED MODULE. DETAIL REPRESENTS THE DETAIL BRANCH. SEMANTIC REPRESENTS THE SEMANTIC BRANCH. AGGREGATION INDICATES THE FUSION STRUCTURE PART

Detail	Semantic		Aggregation			Enhance	mIoU(%)	Mean F1(%)	OA(%)
CBS	RN-18	LMA	CAT	FFM	SI	TTA			
✓							64.30	77.39	82.26
	✓						73.67	83.87	87.79
	✓	✓					75.82	85.68	89.40
✓	✓	✓	✓				79.71	88.51	89.88
✓	✓	✓		✓			80.59	89.09	89.97
✓	✓	✓		✓	✓		81.22	89.48	90.49
✓	✓	✓		✓	✓	✓	**82.03**	**89.99**	**90.88**

TABLE II

ABLATION EXPERIMENTS OF RESIDUAL BRANCH FOR LMA MODULE

Method	mIoU(%)	Mean F1(%)	OA(%)	FPS	Parameters
w/o RB	80.54	89.06	90.19	**117**	13.24M
w/ RB	**81.22**	**89.48**	**90.49**	116	13.24M

TABLE III

ABLATION EXPERIMENTS OF SIS

Method	mIoU(%)	Mean F1(%)	OA(%)	FPS	Parameters
w/o SI	80.59	89.09	89.88	**120**	13.20M
w/ SI-Sem	80.77	89.20	90.26	118	13.24M
w/ SI-Det	80.78	89.22	90.15	119	13.20M
w/ SI	**81.22**	**89.48**	**90.49**	116	13.24M

where N is the number of samples, C is the number of categories, y_{ij} denotes the label (0 or 1) of sample i belonging to category j, and p_{ij} is the probability value of the model output, indicating the predicted probability of sample i belonging to category j.

C. Evaluation Metrics

To evaluate the performance of the network, we used $F1$-score ($F1$), mean intersection over union (mIoU), and overall accuracy (OA) to calculate the metrics of the model

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{15}$$

$$\mathrm{mIoU} = \frac{1}{\mathrm{k}+1} \sum_{\mathrm{i}=0}^{\mathrm{k}} \frac{\mathrm{TP_i}}{\mathrm{FN_i} + \mathrm{FP_i} + \mathrm{TP_i}} \tag{16}$$

$$F1 = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{17}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{18}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{19}$$

where TP is the true predicted positive class sample, TN is the true predicted negative sample, FP is the false predicted positive sample, and FN is the false predicted negative sample. In addition, we use the FPS to evaluate the operation speed of the network.

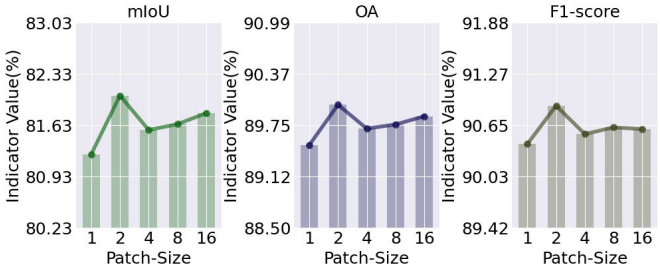


Fig. 6. Experimental setting of the patch size, where the patch size is set to [1, 2, 4, 8, 16] and the three plots correspond to the values of mIoU, $F1$-score, and OA on the Vaihingen test set.

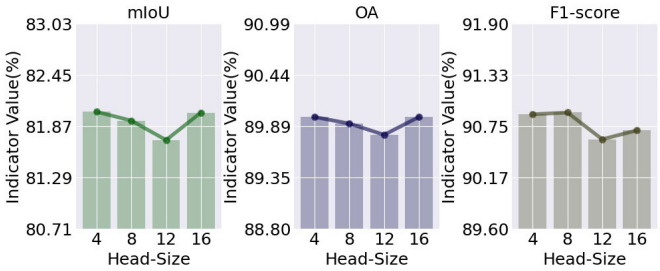


Fig. 7. Experimental setting of number of multiheads, where the head size is set to [4, 8, 12, 16] and the three plots correspond to the values of mIoU, $F1$-score, and OA on the Vaihingen test set.

D. Ablation Study

To evaluate the effectiveness of our method, in this section, we perform an exploration of ablation experiments on the Vaihingen dataset. In addition, we perform experimental validation of some parameter settings of the module.

1) Single Path: We experimented with a single detail branch and a single semantic branch, as shown in the first row and the second/third row of Table I showing the performance of using a single detail branch and a semantic branch, respectively. It can be seen that the model performance is still low when only a single detail branch or semantic branch is used. For the detail branch, it lacks the guidance of high-level semantic information; for the semantic branch, it is complemented by low-level detail information. In the fourth row of Table I, we introduce only a concatenate operation after completing the aggregated splicing of Stage 3 and Stage 4, with a simple coupling of these two branches, and we can see that the

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TABLE IV
COMPARISON WITH STATE-OF-THE-ART LIGHTWEIGHT SEMANTIC SEGMENTATION METHODS FOR REMOTE SENSING IMAGES ON THE VAIHINGEN TEST SET

Method	Backbone	ImSurf	Building	LowVeg	Tree	Car	mIoU(%)	Mean F1(%)	OA(%)
SegNet [15]	-	89.66	91.71	81.84	88.86	67.03	73.12	83.82	87.80
FastSCNN [52]	-	89.80	91.97	80.89	87.86	67.31	72.49	83.40	87.55
CGNet [61]	-	90.90	93.43	82.57	88.68	77.37	76.81	86.59	88.97
BiSeNet [24]	ResNet-18	91.31	94.74	82.27	88.69	78.30	77.59	87.07	89.47
BiSeNetv2 [51]	-	91.59	94.16	82.16	89.01	78.98	77.72	87.18	89.42
PSPNet [17]	ResNet-18	91.17	94.64	82.92	88.99	73.41	76.51	86.23	89.49
DeepLabV3+ [19]	ResNet-50	91.91	94.85	83.21	89.14	86.21	80.54	89.07	89.97
EaNet [62]	ResNet-18	92.00	94.33	83.18	88.93	81.19	78.81	87.92	89.78
DANet [27]	ResNet-18	91.85	94.93	83.81	89.60	80.27	79.12	88.09	90.15
MAResU-Net [54]	ResNet-18	92.52	95.33	**84.72**	90.06	86.64	81.80	89.86	90.78
PSANet [28]	ResNet-50	92.51	95.51	83.31	89.40	87.82	81.60	89.71	90.45
MANet [56]	ResNet-18	92.39	95.15	84.17	**90.12**	**87.71**	81.89	89.91	90.65
ESDINet (Ours)	ResNet-18	**92.74**	**95.54**	84.46	90.03	87.16	**82.03**	**89.99**	**90.88**

TABLE V
COMPETITION WITH THE LIGHTWEIGHT SEMANTIC SEGMENTATION METHODS ON THE VAIHINGEN TEST SET IN TERMS OF PERFORMANCE SPEED

Method	mIoU(%)	Mean F1(%)	OA(%)	FPS	Parameters
ENet [20]	73.44	84.09	88.02	65	**0.35M**
FastSCNN [52]	72.49	83.40	87.55	171	1.13M
BiSeNet [24]	77.59	87.07	89.47	**218**	12.44M
SegNet [15]	73.12	83.82	87.80	39	24.45M
EaNet [62]	78.81	87.92	89.78	125	34.76M
DANet [27]	79.12	88.09	90.15	139	12.63M
MAResU-Net [54]	81.80	89.86	90.78	89	16.17M
MANet [56]	81.89	89.91	90.65	118	12.00M
ESDINet (Ours)	**82.03**	**89.99**	**90.88**	116	13.24M

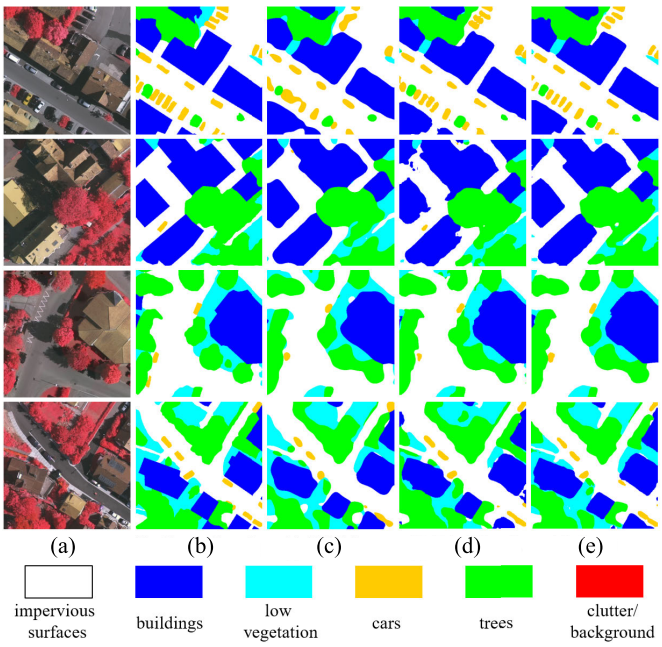


Fig. 8. Visualization results of Vaihingen segmentation for several lightweight models on a cropped dataset, where the image size is 512 × 512. (a) Input. (b) Ground truth. (c) PSPNet. (d) BiseNetv2. (e) Ours.

overall mIoU of the model improves to 79.71%, and the OA and $F1$-score also bring some degree of gain. This indicates that the characteristic information of these two branches is complementary.

2) Semantic Branch: We compared and validated the hybrid structure of ResNet-18 and the LMA introduced in the semantic branch, respectively, as shown in the second and third rows of Table I. The introduction of LMA improves the ability of the semantic branch to capture contextual information, which makes the hybrid structure have both local and global perceptual properties, improves the detection accuracy of small object targets (especially the car class), and contributes significantly to the accuracy improvement of the whole model. Compared with the initial pure convolutional structure, the overall mIoU of the model is significantly improved by 2.15% (73.67% → 75.82%), $F1$-score by 1.81% (83.87% → 85.68%), and OA by 1.61% (87.79% → 89.40%) using the hybrid structure with the introduction of LMA. For the whole model, we verified the impact of introducing a residual branch. The residual branch is able to provide local information to aid linear attention, making the LMA locally and globally information aware. As shown in Table II, LMA achieves better results with the introduction of residual path aids.

3) Feature Fusion Module: In the fusion module, we first replace the concatenate operation in deep aggregation with our proposed FFM module, as shown in the fifth line. After introducing FFM, the model is more sensitive to the important information in the high-level features of the two branches and can effectively fuse the detailed features of the detail branch with the semantic features of the semantic branch. The comparison reveals that the use of the FFM module improves

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TABLE VI
COMPARISON WITH STATE-OF-THE-ART LIGHTWEIGHT SEMANTIC SEGMENTATION METHODS FOR REMOTE SENSING IMAGES ON THE POTSDAM TEST SET

Method	Backbone	ImSurf	Building	LowVeg	Tree	Car	mIoU(%)	Mean F1(%)	OA(%)
SegNet [15]	-	89.83	93.27	84.62	84.37	92.51	80.26	88.92	87.21
FastSCNN [52]	-	90.45	94.48	85.35	85.27	91.84	81.15	89.48	88.29
CGNet [61]	-	90.31	94.51	84.81	85.67	92.37	81.25	89.53	88.12
BiSeNet [24]	ResNet-18	91.07	95.12	85.81	86.63	91.37	81.99	90.00	89.06
BiSeNetv2 [51]	-	90.92	94.92	85.99	86.17	91.93	81.97	89.99	89.98
PSPNet [17]	ResNet-18	90.79	95.15	86.57	87.32	91.05	82.25	90.18	89.31
DeepLabV3+ [19]	ResNet-50	91.89	95.35	86.68	87.68	93.94	83.84	91.11	89.80
EaNet [62]	ResNet-18	91.62	95.27	85.83	86.89	93.45	83.04	90.61	89.15
DANet [27]	ResNet-18	92.31	96.21	86.82	88.14	94.85	84.83	91.67	90.23
MAResU-Net [54]	ResNet-18	92.58	96.15	86.68	88.22	**95.59**	85.15	91.84	90.36
PSANet [28]	ResNet-50	92.49	96.28	**87.31**	**88.61**	95.58	**85.48**	**92.05**	**90.55**
MANet [56]	ResNet-18	92.36	96.13	87.00	88.04	95.34	85.01	91.77	90.34
ESDINet (Ours)	ResNet-18	**92.68**	**96.28**	87.29	88.10	95.42	85.32	91.96	90.53

TABLE VII
COMPETITION WITH THE LIGHTWEIGHT SEMANTIC SEGMENTATION METHODS ON THE POTSDAM TEST SET IN TERMS OF PERFORMANCE SPEED

Method	mIoU(%)	Mean F1(%)	OA(%)	FPS	Parameters
ENet [20]	80.91	89.28	88.00	65	**0.35M**
FastSCNN [52]	81.15	89.48	88.29	171	1.13M
BiSeNet [24]	81.99	90.00	89.06	**218**	12.44M
SegNet [15]	80.26	88.92	87.21	39	24.45M
EaNet [62]	83.04	90.61	89.15	125	34.76M
DANet [27]	84.83	91.67	90.23	139	12.63M
MAResU-Net [54]	85.15	91.84	90.36	89	16.17M
MANet [56]	85.01	91.77	90.34	118	12.00M
ESDINet (Ours)	**85.32**	**91.96**	**90.53**	116	13.24M

mIoU by 0.88% (79.71% → 80.59%), $F1$-score by 0.58% (88.51% → 89.09%), and OA by 0.09% (89.88% → 89.97%).

4) Shallow Interaction: We explore the adopted SI module, i.e., in addition to using FFM for interaction in the deep part of the network, we also design the corresponding interaction structure in the shallow part of the network. The multiscale aggregation of high-level semantics and low-level details is achieved by making full use of the complementary information properties of the two-branch structure. Also, due to the simple and efficient structure adopted for the shallow part, the entire SI introduces an almost negligible number of parameters and computational effort. As can be seen from the fifth and sixth rows of Table I, the additional inclusion of SI improves the model mIoU by 0.63% (80.59% → 81.22%), $F1$-score by 0.39% (89.09% → 89.48%), and OA by 0.52% (89.97% → 90.49%). This indicates that the shallow information of the detail branch and the semantic branch can be exploited to complementary effect.

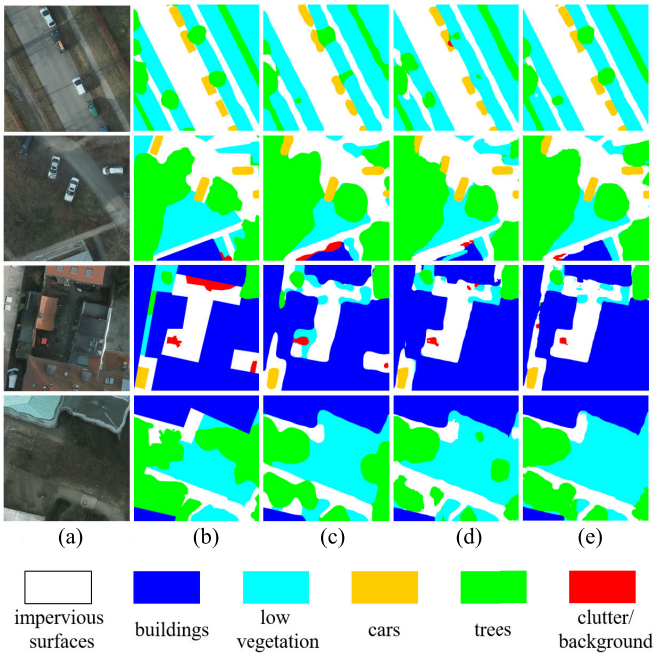


Fig. 9. Visualization results of Potsdam segmentation for several lightweight models on a cropped dataset, where the image size is 512 × 512. (a) Input. (b) Ground truth. (c) PSPNet. (d) BiseNetv2. (e) Ours.

More detailed experiments on SIs are presented in Table III. For example, in the second and third rows, we ablate separate guidance for detail and semantic branches in the shallow layer of the network, respectively. As can be seen, in the shallow layer of the network, the overall performance of the model is improved when the semantic and detail branches are guided by one-way spatial details (second row of Table III) and one-way high-level semantics (third row of Table III), respectively, indicating that both semantic and detail information in the shallow layer can provide auxiliary information guidance. And, the overall gain of the model is most effective when

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TABLE VIII
COMPARISON WITH STATE-OF-THE-ART LIGHTWEIGHT SEMANTIC SEGMENTATION METHODS FOR REMOTE SENSING IMAGES ON THE LOVEDA TEST SET

Method	Backbone	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU(%)
ENet [20]	-	38.15	49.76	50.03	74.12	14.26	41.94	49.75	45.43
CGNet [61]	-	38.65	48.07	52.11	74.73	16.89	40.81	48.10	45.63
BiSeNet [24]	ResNet-18	41.78	50.23	52.54	74.93	14.05	41.99	56.12	47.37
BiSeNetv2 [51]	-	43.04	51.19	52.77	75.93	14.99	43.66	**60.65**	48.89
PSPNet [17]	ResNet-50	44.15	54.72	56.54	76.81	17.49	**47.13**	57.54	**50.63**
DeepLabV3+ [19]	ResNet-50	43.35	54.40	**56.96**	76.61	17.65	47.08	57.69	50.54
MAResU-Net [54]	ResNet-18	**48.35**	42.75	53.56	77.12	15.26	44.77	56.53	48.35
EaNet [62]	ResNet-18	38.69	51.67	52.16	75.27	18.36	40.59	54.11	47.26
MANet [54]	ResNet-18	43.29	**55.35**	56.20	74.18	14.28	42.85	58.62	49.26
ESDINet (Ours)	ResNet-18	41.57	53.77	54.84	**78.71**	**19.49**	44.17	57.95	50.07

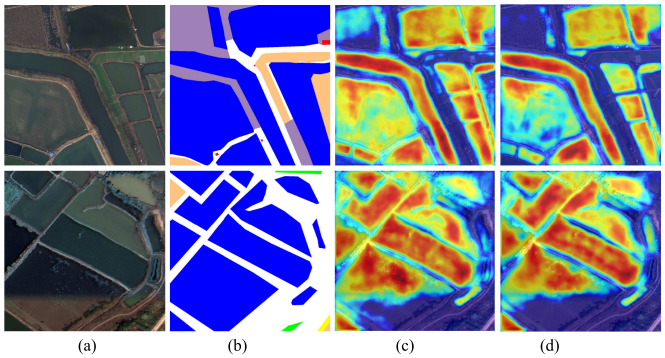


Fig. 10. Attention heatmap visualization of LoveDA Dataset (water category), corresponding to different settings without and with LMA, respectively. (a) Input. (b) Ground truth. (c) w/o LMA. (d) w/ LMA.

both branches use the interaction structure (i.e., using the SI module).

5) Parameter Analysis: We also conducted exploratory experiments for the patch size and the number of multiheads of LMA to verify the reasonableness of the parameter settings. In the patch size setting experiment, we constructed five different groups of sizes [1, 2, 4, 8, 16], as shown in Fig. 6. The results show that the best results are obtained when the patch size is 2. From the size of the feature map and the high-level information it contains, it is intuitively easier to capture some more detailed informative features with patch sizes set to 2, since we introduce the LMA module only at the later stage of semantic branching. (The sizes of high-level features are 32×32 and 16×16, respectively.)

In the experiments of multiheads number setting, we constructed four sets of models with increasing multiheads numbers of [4, 8, 12, 16], as shown in Fig. 7. It can be seen that the difference of the multihead number on the overall model is relatively small, where the accuracy achieves better results when the multiheads number is set to 4 and 16. Considering that, the resolution of the feature map is small in the late semantic branching, using a small number of multiheads can already achieve a better subspace mapping, so we take the setting of the number of heads as 4.

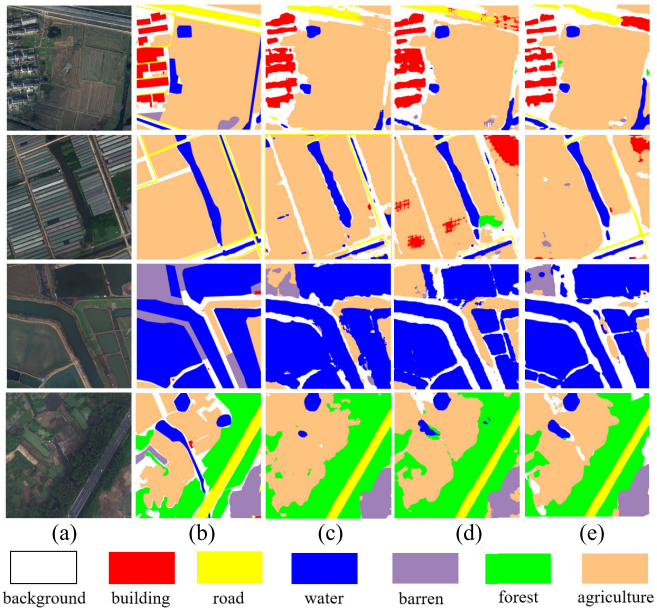


Fig. 11. Visualization results of LoveDA segmentation for several lightweight models on a cropped dataset, where the image size is 1024×1024. (a) Input. (b) Ground truth. (c) PSPNet. (d) MANet. (e) Ours.

E. Result

In this section, we experimentally compare our ESDINet with other state-of-the-art methods on three benchmarks. The inference measurements of FPS were also performed on a computer with a GPU of 2080 Ti.

1) Results on the Vaihingen Dataset: The evaluation results of the existing state-of-the-art methods on the Vaihingen test set are shown in Table IV. From the test results in Table IV, it is clear that ESDINet, as a lightweight model, is also able to achieve competitive experimental results compared with other semantic segmentation models of complex design.

In addition, we also measure the comparative segmentation accuracy and inference speed of the lightweight model on the pairwise Vaihingen test set. As shown in Table V, ESDINet showed competitive results in mIoU, $F1$-score, and OA. The shallow-deep interaction structure we construct, combined with LMA to enhance global information extraction, enables

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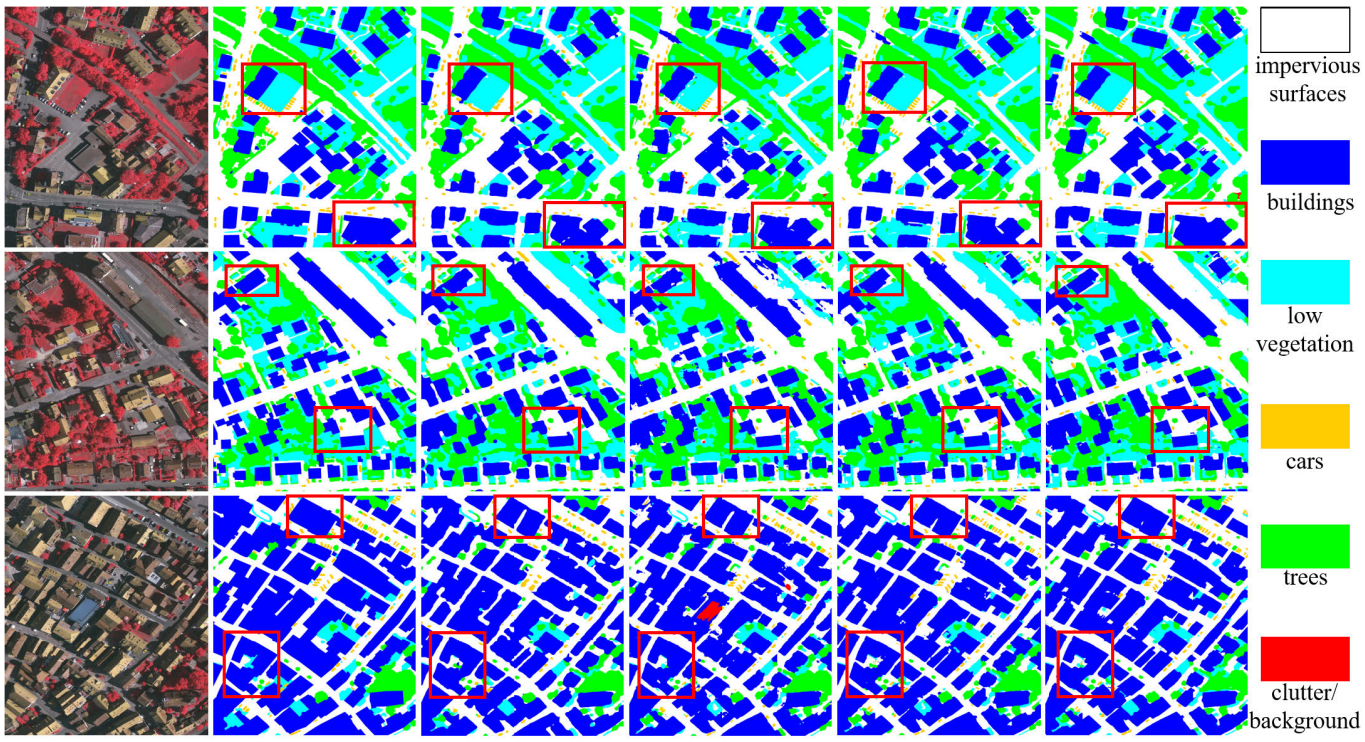


Fig. 12. Segmentation results for the whole image on the Vaihingen test set. The red boxes mark some distinctly different segmentation regions. (a) Input. (b) Ground truth. (c) PSPNet. (d) BiseNetv2. (e) DeepLabV3+. (f) Ours.

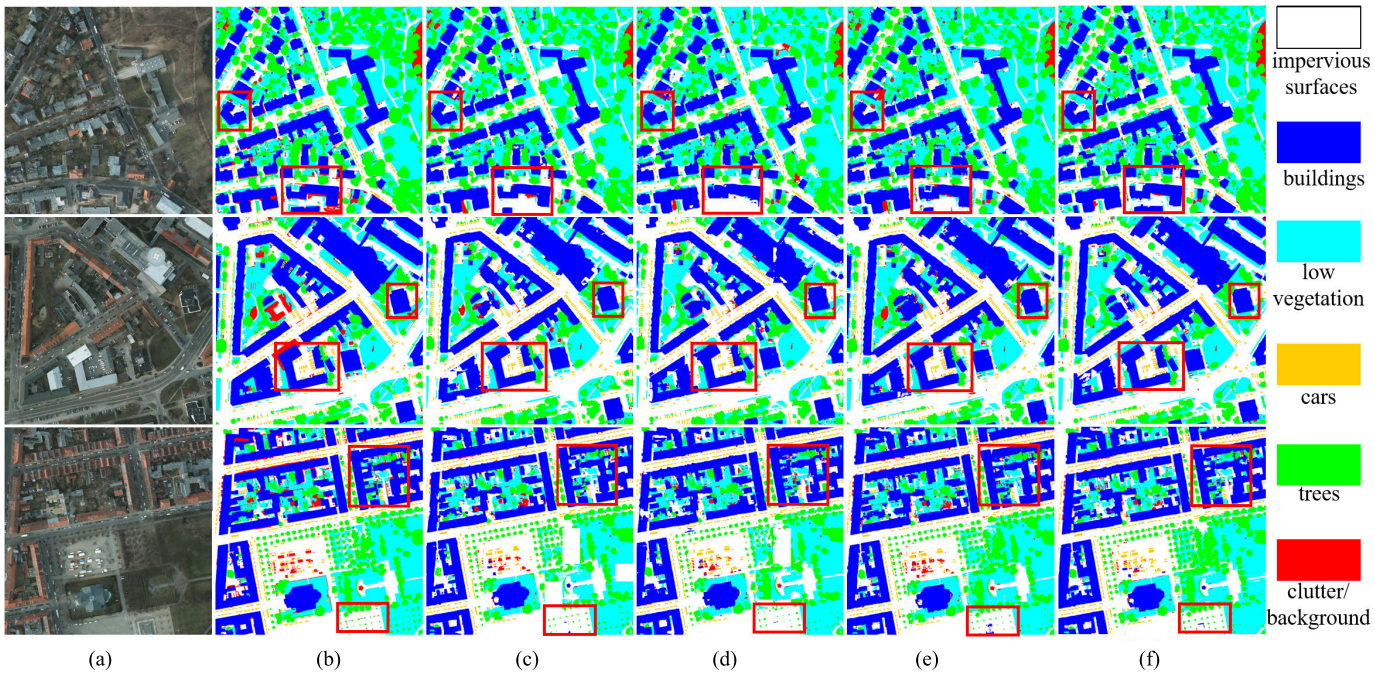


Fig. 13. Segmentation results for the whole image on the Potsdam test set. The red boxes mark some distinctly different segmentation regions. (a) Input. (b) Ground truth. (c) PSPNet. (d) BiseNetv2. (e) DeepLabV3+. (f) Ours.

fuller exploitation of the multiscale feature representation of both branches. Also, since we designed the structure to be as simple as possible and used some efficient modules, our whole model is lightweight, with fewer parameters and faster inference. It can be seen that our method achieves high results for both the building and car objects, where the building achieves an $F1$-score of 95.54%, which is higher than the performance of the other models.

Fig. 8 shows some of our segmentation visualization results in the local cut, and Fig. 12 shows the segmentation visualization results for the whole image. Visual comparisons indicate that our approach exhibits finer handling of target

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objects, particularly in distinguishing the edges of the car class, in contrast to BiSeNet. To a significant extent, the diverse interaction integration between low-level and high-level features employed in our model contributes to improving this situation.

2) Results on the Potsdam Dataset: Our experimental results for the Potsdam test set are shown in Table VI. Compared with the Vaihingen dataset, the Potsdam dataset has a higher resolution. Table VII shows the segmentation accuracy and inference speed of the lightweight model on a pair of Potsdam test sets. The results show that ESDINet is able to achieve a good balance between speed and accuracy, where the $F1$-score of car is 95.42%, showing that our model has a good perception of small target objects. The visualization results for specific regions are shown in Fig. 9, and the segmentation results for the whole image are shown in Fig. 13. Based on the results, our ESDINet achieves greater accuracy in the recognition of classes such as ImSurf, building, and LowVeg, which tend to appear more frequently. These characteristics contribute to our model efficiently integrating detailed and semantic information through various interaction mechanisms and attention modules, thereby leading to improvements in edge segmentation.

3) Results on the LoveDA Dataset: Our experimental results for the LoveDA test set are shown in Table VIII. As can be seen from the results in the table, ESDINet is able to show competitive performance (50.07% mIoU) with a lightweight model. Among these, ESDINet demonstrates excellent performance in the water and barren categories, while its performance in the background category is slightly lower. Fig. 10 illustrates the impact of different settings of LMA on attention heatmaps. It can be observed that with the introduction of LMA, the model is capable of capturing semantic features of the interested regions in a more refined manner. Fig. 11 displays the visual results on the LoveDA validation set. Visual results indicate that our model performs well in the water category, while its performance is slightly lower in some background classes. As shown in the third row of Fig. 11, our model incorrectly identifies many categories as background. For data with complex scenarios (situations where category boundaries overlap), additional learning is required after introducing the Transformer structure in our model to fit better and produce improved results. Overall, compared with the more computationally intensive network models, ESDINet is able to achieve relatively good performance in the face of more complex data.

V. Conclusion

In this article, we propose a new lightweight two-branch network structure, including a detail branch for extracting spatial details and a semantic branch for obtaining the global context, to address the accuracy shortage of existing lightweight networks in high-resolution remote sensing image segmentation. Different interaction structures of the shallow (SI) and deep (FFM) layers enable a fast and effective fusion of the two branches of information. In addition, we designed a linear attention-based LMA module and introduced a hybrid structure of constructed CNN and Transformer in the semantic branch. We conducted extensive experimental and visualization explorations for the proposed method. The results show that our improvements enhance the information feature representation of the model and effectively improve the overall segmentation performance of the network while ensuring real-time inference speed. In the future, we will further improve ESDINet in the following aspects: 1) optimize the more frequently used convolution operations and explore how to maintain a small computational cost while enhancing local feature extraction performance and 2) explore more optimal auxiliary enhancement strategies to address the underperformance of linear attention.

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